**SECTION 1 (Maximum Marks: 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : $+3$ If **ONLY** the correct option is chosen;
*Zero Marks* : $0$ If none of the options is chosen (i.e. the question is unanswered);
*Negative Marks* : $-1$ In all other cases.

Q.1 Considering only the principal values of the inverse trigonometric functions, the value of

$$\tan\left(\sin^{-1}\left(\frac{3}{5}\right)-2\cos^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$$

is

(A) $\frac{7}{24}$ (B) $\frac{-7}{24}$

(C) $\frac{-5}{24}$ (D) $\frac{5}{24}$

Q.2 Let $S=\left\{(x, y) \in \mathbb{R} \times \mathbb{R}: x \geq 0, y \geq 0, y^2 \leq 4x, y^2 \leq 12-2x \text{ and } 3y+\sqrt{8}\, x \leq 5\sqrt{8}\right\}$. If the area of the region $S$ is $\alpha\sqrt{2}$, then $\alpha$ is equal to

(A) $\frac{17}{2}$ (B) $\frac{17}{3}$ (C) $\frac{17}{4}$ (D) $\frac{17}{5}$

Q.3 Let $k \in \mathbb{R}$. If $\lim\limits_{x \to 0+} \left(\sin(\sin kx)+\cos x+x\right)^{\frac{2}{x}}=e^6$, then the value of $k$ is

(A) 1 (B) 2 (C) 3 (D) 4

Q.4 Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by

$$f(x)=\begin{cases} x^2\sin\left(\dfrac{\pi}{x^2}\right), & \text{if } x\neq 0, \\ 0, & \text{if } x=0. \end{cases}$$

Then which of the following statements is TRUE?

(A) $f(x)=0$ has infinitely many solutions in the interval $\left[\dfrac{1}{10^{10}},\ \infty\right)$.

(B) $f(x)=0$ has no solutions in the interval $\left[\dfrac{1}{\pi},\ \infty\right)$.

(C) The set of solutions of $f(x)=0$ in the interval $\left(0,\ \dfrac{1}{10^{10}}\right)$ is finite.

(D) $f(x)=0$ has more than 25 solutions in the interval $\left(\dfrac{1}{\pi^2},\ \dfrac{1}{\pi}\right)$.

**SECTION 2 (Maximum Marks: 12)**

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme</u>:

  *Full Marks* : $+4$ **ONLY** if (all) the correct option(s) is(are) chosen;
  *Partial Marks* : $+3$ If all the four options are correct but **ONLY** three options are chosen;
  *Partial Marks* : $+2$ If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  *Partial Marks* : $+1$ If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  *Zero Marks* : $0$ If unanswered;
  *Negative Marks :* $-2$ In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get $+4$ marks;
  choosing ONLY (A) and (B) will get $+2$ marks;
  choosing ONLY (A) and (D) will get $+2$marks;
  choosing ONLY (B) and (D) will get $+2$ marks;
  choosing ONLY (A) will get $+1$ mark;
  choosing ONLY (B) will get $+1$ mark;
  choosing ONLY (D) will get $+1$ mark;
  choosing no option(s) (i.e. the question is unanswered) will get $0$ marks and
  choosing any other option(s) will get $-2$ marks.

Q.5 Let $S$ be the set of all $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$ such that

$$\lim_{x \to \infty} \frac{\sin(x^2)(\log_e x)^{\alpha} \sin\left(\frac{1}{x^2}\right)}{x^{\alpha\beta}(\log_e(1+x))^{\beta}} = 0.$$

Then which of the following is (are) correct?

(A) $(-1,\ 3) \in S$
(B) $(-1,\ 1) \in S$
(C) $(1,\ -1) \in S$
(D) $(1,\ -2) \in S$

Q.6 A straight line drawn from the point $P(1,3,2)$, parallel to the line $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-6}{1}$, intersects the plane $L_1: x-y+3z=6$ at the point $Q$. Another straight line which passes through $Q$ and is perpendicular to the plane $L_1$ intersects the plane $L_2: 2x-y+z=-4$ at the point $R$. Then which of the following statements is (are) TRUE?

(A) The length of the line segment $PQ$ is $\sqrt{6}$
(B) The coordinates of $R$ are $(1,6,3)$
(C) The centroid of the triangle $PQR$ is $\left(\frac{4}{3},\frac{14}{3},\frac{5}{3}\right)$
(D) The perimeter of the triangle $PQR$ is $\sqrt{2}+\sqrt{6}+\sqrt{11}$

Q.7 Let $A_1, B_1, C_1$ be three points in the $xy$-plane. Suppose that the lines $A_1C_1$ and $B_1C_1$ are tangents to the curve $y^2=8x$ at $A_1$ and $B_1$, respectively. If $O=(0,0)$ and $C_1=(-4,0)$, then which of the following statements is (are) TRUE?

(A) The length of the line segment $OA_1$ is $4\sqrt{3}$
(B) The length of the line segment $A_1B_1$ is 16
(C) The orthocenter of the triangle $A_1B_1C_1$ is $(0,0)$
(D) The orthocenter of the triangle $A_1B_1C_1$ is $(1,0)$

**SECTION 3 (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : $+4$ If **ONLY** the correct integer is entered;
  *Zero Marks* : $0$ In all other cases.

Q.8 Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x+y)=f(x)+f(y)$ for all $x,\ y\in\mathbb{R}$, and $g:\mathbb{R}\to(0,\infty)$ be a function such that $g(x+y)=g(x)g(y)$ for all $x, y\in\mathbb{R}$. If $f\left(\frac{-3}{5}\right)=12$ and $g\left(\frac{-1}{3}\right)=2$, then the value of $\left(f\left(\frac{1}{4}\right)+g(-2)\ -8\right)g(0)$ is _____.

Q.9 A bag contains $N$ balls out of which $3$ balls are white, $6$ balls are green, and the remaining balls are blue. Assume that the balls are identical otherwise. Three balls are drawn randomly one after the other without replacement. For $i=1,2,3$, let $W_i, G_i,$ and $B_i$ denote the events that the ball drawn in the $i^{\text{th}}$ draw is a white ball, green ball, and blue ball, respectively. If the probability $P(W_1\cap G_2\cap B_3)=\frac{2}{5N}$ and the conditional probability $P(B_3\mid W_1\cap G_2)=\frac{2}{9}$, then $N$ equals _____.

Q.10 Let the function $f:\mathbb{R}\to\mathbb{R}$ be defined by

$$f(x)=\frac{\sin x}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}+\frac{2}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}.$$

Then the number of solutions of $f(x)=0$ in $\mathbb{R}$ is _____.

Q.11 Let $\vec{p}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{q}=\hat{i}-\hat{j}+\hat{k}$. If for some real numbers $\alpha, \beta,$ and $\gamma,$ we have

$$15\hat{i}+10\hat{j}+6\hat{k}=\alpha\left(2\vec{p}+\vec{q}\right)+\beta\left(\vec{p}-2\vec{q}\right)+\gamma\left(\vec{p}\times\vec{q}\right),$$

then the value of $\gamma$ is _____.

Q.12 A normal with slope $\frac{1}{\sqrt{6}}$ is drawn from the point $(0,-\alpha)$ to the parabola $x^2=-4ay,$ where $a>0.$ Let $L$ be the line passing through $(0,-\alpha)$ and parallel to the directrix of the parabola. Suppose that $L$ intersects the parabola at two points $A$ and $B.$ Let $r$ denote the length of the latus rectum and $s$ denote the square of the length of the line segment $AB.$ If $r:s=1:16,$ then the value of $24a$ is _____.

Q.13 Let the function $f:[1,\infty)\to\mathbb{R}$ be defined by

$$f(t)=\begin{cases}(-1)^{n+1}2, & \text{if } t=2n-1,\ n\in\mathbb{N},\\ \frac{(2n+1-t)}{2}f(2n-1)+\frac{(t-(2n-1))}{2}f(2n+1), & \text{if } 2n-1<t<2n+1,\ n\in\mathbb{N}.\end{cases}$$

Define $g(x)=\int_1^x f(t)\,dt,\ x\in(1,\infty).$ Let $\alpha$ denote the number of solutions of the equation $g(x)=0$ in the interval $(1,8]$ and $\beta=\lim\limits_{x\to1+}\frac{g(x)}{x-1}.$ Then the value of $\alpha+\beta$ is equal to _____.

**SECTION 4 (Maximum Marks: 12)**

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, there are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

  *Full Marks* : $+3$ If ONLY the correct numerical value is entered in the designated place;
  *Zero Marks* : $0$ In all other cases.

**PARAGRAPH "I"**

Let $S=\{1,2,3,4,5,6\}$ and $X$ be the set of all relations $R$ from $S$ to $S$ that satisfy both the following properties:

i. $R$ has exactly 6 elements.
ii. For each $(a,b)\in R$, we have $|a-b|\geq 2$.

Let $Y=\{R\in X:$ The range of $R$ has exactly one element$\}$ and
$Z=\{R\in X: R$ is a function from $S$ to $S\}$.
Let $n(A)$ denote the number of elements in a set $A$.

**(There are two questions based on PARAGRAPH "I", the question given below is one of them)**

Q.14 If $n(X)={}^{m}C_{6}$, then the value of $m$ is _____.

**PARAGRAPH "I"**

Let $S=\{1,2,3,4,5,6\}$ and $X$ be the set of all relations $R$ from $S$ to $S$ that satisfy both the following properties:

i. $R$ has exactly 6 elements.
ii. For each $(a,b)\in R$, we have $|a-b|\geq 2$.

Let $Y=\{R\in X:$ The range of $R$ has exactly one element$\}$ and
$Z=\{R\in X: R$ is a function from $S$ to $S\}$.
Let $n(A)$ denote the number of elements in a set $A$.

**(There are two questions based on PARAGRAPH "I", the question given below is one of them)**

Q.15 If the value of $n(Y)+n(Z)$ is $k^2$, then $|k|$ is _____.

**PARAGRAPH "II"**

Let $f:\left[0,\frac{\pi}{2}\right]\to[0,1]$ be the function defined by $f(x)=\sin^2 x$ and let $g:\left[0,\frac{\pi}{2}\right]\to[0,\infty)$ be the function defined by $g(x)=\sqrt{\frac{\pi x}{2}-x^2}$.

**(There are two questions based on PARAGRAPH "II", the question given below is one of them)**

Q.16 The value of $2\int_0^{\frac{\pi}{2}} f(x)g(x)dx-\int_0^{\frac{\pi}{2}} g(x)dx$ is _____.

**PARAGRAPH "II"**

Let $f:\left[0,\frac{\pi}{2}\right]\to[0,1]$ be the function defined by $f(x)=\sin^2 x$ and let $g:\left[0,\frac{\pi}{2}\right]\to[0,\infty)$ be the function defined by $g(x)=\sqrt{\frac{\pi x}{2}-x^2}$.

**(There are two questions based on PARAGRAPH "II", the question given below is one of them)**

Q.17 The value of $\frac{16}{\pi^3}\int_0^{\frac{\pi}{2}} f(x)g(x)dx$ is _____.

## END OF THE QUESTION PAPER